

Knative: Serverless Computing in Partnership with Kubernetes

Knative adds high-level features to Kubernetes that make serverless computing so much easier.

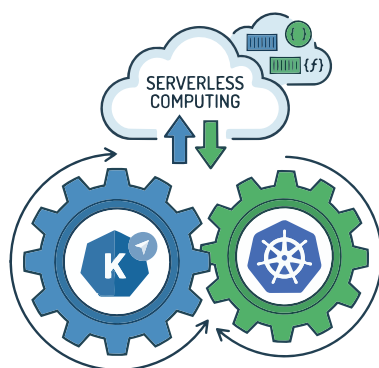
Serverless computing has changed the way we deploy and scale modern apps. Developers can run code on platforms like AWS Lambda without having to worry about servers. This speeds up development cycles and lets the code automatically scale. But many companies quickly run into a big problem: vendor lock-in. When apps are tightly linked to a private cloud provider, it can be hard and expensive to move workloads or use a multi-cloud strategy.

Knative is one of the most important things that has happened in this field. Built on top of Kubernetes, it offers a way to do serverless computing that is native to Kubernetes. Instead of hiding containers completely, it adds high-level features to Kubernetes, like automatic scaling, event-driven workloads, and easier deployment of stateless services.

What is Knative?

Kubernetes does not need to be replaced by Knative. Instead, the latter adds features to Kubernetes that make it easier to run applications that don't need servers and are event-driven. It adds more parts that take care of scaling, routing, and event management, but it still uses Kubernetes for the main container orchestration. The main parts are:

Serving: This part oversees deploying and managing workloads that don't need a server. It automatically takes care of scaling, deploying containers, and routing traffic. One of its best features is scale-to-zero, which



means that inactive services don't use any resources until a request comes in.

Eventing: Knative Eventing is a way to make architectures that are based on events. It lets services talk to each other through events instead of direct requests. Developers can link event producers, brokers, and consumers in flexible pipelines, which makes it possible for workflows like processing data in real time, messaging systems, and orchestrating microservices.

Build (historical background):

In earlier versions of Knative, there was a Build component that made it easier to create container images right in the platform. But this feature was eventually split and turned into separate tools like Tekton. CI/CD pipelines are usually managed by someone else these days, while Knative focuses on serving and running events.

Model of the architecture

Knative's architectural model is meant to add serverless features to containerised environments while keeping Kubernetes's flexibility. Knative doesn't

depend on proprietary infrastructure; instead, it adds a set of abstractions that make it easier to deploy, scale, and manage traffic for modern apps.

Autoscaling to 0: Another important feature is the ability to automatically scale down to zero. When Kubernetes runs traditional containerised services, they often use resources even when they aren't doing anything. When no requests are being processed, Knative automatically scales services down to zero instances to fix this.

Dividing traffic: Knative also has advanced tools for managing traffic. Developers can send traffic to different versions of a service, which lets them roll out changes slowly and use safer deployment methods. For instance, a new version of an app might get a small amount of traffic while the stable version handles most of the requests.

Workloads that are event-driven:

Knative's eventing system lets you run event-driven workloads in addition to request-based ones. Apps can react to events that come from different places, like messaging systems, data streams, or other services.

Why teams use Knative in production

Deployments in hybrid and multi-cloud:

Knative is especially useful for teams that work in environments with more than one cloud. It can be run on Kubernetes, which means that the same application architecture can be used on different cloud providers or even in private data centres.